

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) Semester III Examination 2025-26 (NEP)

**COMPUTER SCIENCE (SEC)****Paper Code: CSCSE31 : Python Programming**Time:  $2\frac{1}{2}$  Hours

Maximum Marks: 70

*Instructions: All questions are compulsory. Sub-questions to be attempted are indicated in the question itself.*

**Question 1.** Attempt ALL of the following: [2 × 5 = 10]

- What is the difference between a list and a tuple in Python?
- What keyword is used to import a module in Python? Write one import example.
- What is the use of the `__init__()` method in a Python class?
- What does the `open()` function do in file handling? Give its full syntax.
- What is an f-string in Python, and how do you use it to include a variable in a string?

**Question 2.** Attempt any THREE of the following: [4 × 3 = 12]

- Write a Python program to check if a number is positive, negative, or zero.
- Explain how the `=` operator differs from `==` in Python. Provide an example.
- What will be the output of the `random.randint(1, 10)` function? Explain.
- Explain the concept of Abstraction and provide an example in Python.

**Question 3.** Attempt any THREE of the following: [4 × 3 = 12]

- Define Inheritance in Python. Write a simple example to show how a child class can use a method from a parent class.
- Write a Python program that accepts five numbers from the user one by one, finds their sum, and displays whether the sum is even or odd.
- Define a Dictionary in Python and demonstrate how to access, add, and remove elements.
- Explain Polymorphism in Python with a suitable example where the same method name behaves differently for different objects.

**Question 4.** Attempt any THREE of the following: [4 × 3 = 12]

- What is the difference between `open('file', 'w')` and `open('file', 'a')` in Python?
- Differentiate between `remove()` and `pop()` in the context of Python lists with a suitable example.
- What is Encapsulation in OOP? Explain with an example.
- Write a Python program to plot a line graph showing the variation of temperature over a week using Matplotlib.

**Question 5.** Attempt any TWO of the following:

[6 × 2 = 12]

- a. What is programming? Explain the purpose of programming and describe the concepts of Variables, Input/Output (I/O), Operators, and Control Structures with suitable examples.
- b. Write a Python program that uses a recursive function to calculate and display the factorial of a number entered by the user.
- c. Explain the concept of File Handling in Python. Write a Python program to create a file, write some content, append more content, and then read and display the content from the file.

**Question 6.** Attempt any TWO of the following:

[6 × 2 = 12]

- a. Create a Python Bank Account system using Object-Oriented Programming principles. Implement methods for deposit, withdraw, and check balance.
- b. Write a Python program to create and use a user-defined module that calculates the area and perimeter of a circle.
- c. Write a Python program that accepts employee details including name, designation, and salary from the user: Format the data in rows whereby in each row, the name is left-aligned in 30 characters, the designation is left-aligned in 20 characters, and the salary is right-aligned in 10 characters with 2 decimal places. Save the formatted data to a file.

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